MILITARY SPECIFICATION

MOLYBDENUM DISULFIDE, TECHNICAL, LUBRICATION GRADE

This amendment forms a part of Military Specification MIL-M-7866C dated 11 July 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

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Paragraph 4.4.3: Delete and substitute:

"4.4.3 Water solubles. A 10 g sample of the dried molybdenum disulfide shall be weighed into an 80 by 33 mm single thickness thimble. The thimble shall be placed in the extraction tube of a size B Soxhlet extraction apparatus. One hundred and fifty ml of water shall be added to a previously dried and weighed 250 ml flat bottom extraction flask, and the complete extraction apparatus assembled on an electric hotplate. A blank determination shall be run simultaneously. The extraction of the powder and the blank shall be allowed to continue for 4 hours. The tared flasks shall be placed on a steam bath and evaporated to dryness. The flasks containing the residues shall be removed to an oven maintained at 107 $\pm$ 3°C (225 $\pm$ 5°F) for 1 hour, then cooled in a desiccator and weighed.

$$\text{Percent water soluble matter} = \frac{A-B}{W} \times 100$$

Where A = Increase in weight of sample flask
B = Increase in weight of blank flask
W = Weight of dried sample"

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